Joint and conditional distributions

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Summary

The joint distribution of extreme hazards, such as rain, wind, and surge, tells us the probability that they are all large as well as the distribution of one given the others. This session defines joint and conditional distributions in two and then N dimensions, and then asks how to estimate one from data, using kernel density estimates, Bayesian networks, and copulas as examples.

For helpful background, explore:

- Blitzstein & Hwang (2019) §7.1, pp. 304-323, on joint, marginal, and conditional distributions, and §5.3, pp. 224-230, on the probability integral transform.
- Friedman et al. (2009) §6.6.1, pp. 208-210, on kernel density estimation.
- Sebastian et al. (2017) §2, pp. 43-44, for a short introduction to Bayesian networks.
- Genest & Favre (2007), pp. 347-353, on copulas.

Bibliography

Blitzstein, J. K., & Hwang, J. (2019). *Introduction to Probability, Second Edition* (2nd Edition). Boca Raton: Chapman and Hall/CRC. Retrieved from http://probabilitybook.net/

Friedman, J., Hastie, T., & Tibshirani, R. (2009). *The Elements of Statistical Learning* (Vol. 1). Springer series in statistics Springer, Berlin.

Genest, C., & Favre, A.-C. (2007). Everything You Always Wanted to Know about Copula Modeling but Were Afraid to Ask. *Journal of Hydrologic Engineering*, *12*(4), 347–368. https://doi.org/10.1061/(ASCE)1084-0699(2007)12:4(347)

Sebastian, A., Dupuits, E. J. C., & Morales-Nápoles, O. (2017). Applying a Bayesian Network Based on Gaussian Copulas to Model the Hydraulic Boundary Conditions for Hurricane Flood Risk Analysis in a Coastal Watershed. *Coastal Engineering*, *125*, 42–50. https://doi.org/10.1016/j.coastaleng.2017.03.008